

Weld Schedule Manager (Root Pass)



WPS No	J Prep TW1	Position / Direction	Fixed 5G - Down	Diameter (in)	48	Wall (mm)	15.8
Process	LSC (Modified Short Arc)	Specification	API1104	Filler Wire	Surearc ER70S-6(1.1mm)	Shield Gas	Ar/CO2 85/15
Bevel Design	J Prep	Material Grade	X70	Trade Name	Sure-arc	Completed by	Ben Hatch

0°	Travel Speed	Torch Angle	Dynamic Correction
0mm	14	-20	-2
1mm	12	10	-1
2mm	7	10	0
3mm	4.5	10	1
4mm	3	10	2
5mm	2	5	3

30°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15	-15	-2
1mm	13	0	-1
2mm	9	0	0
3mm	6.5	0	0
4mm	4.5	-5	0
5mm	2.75	-5	0

60°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15.5	-10	-2
1mm	13.5	-5	-1
2mm	10	-10	0
3mm	7.5	-15	0
4mm	5	-15	0
5mm	2.75	-15	0

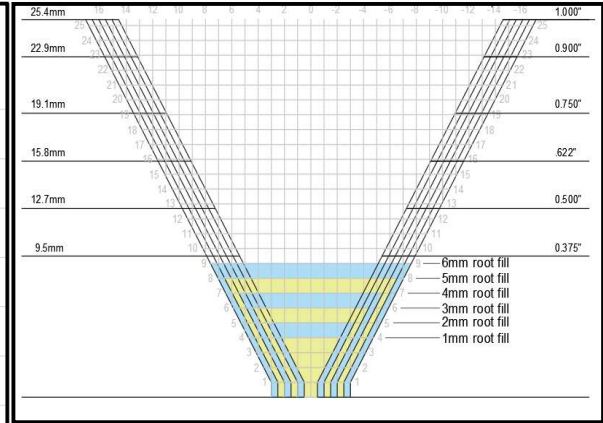
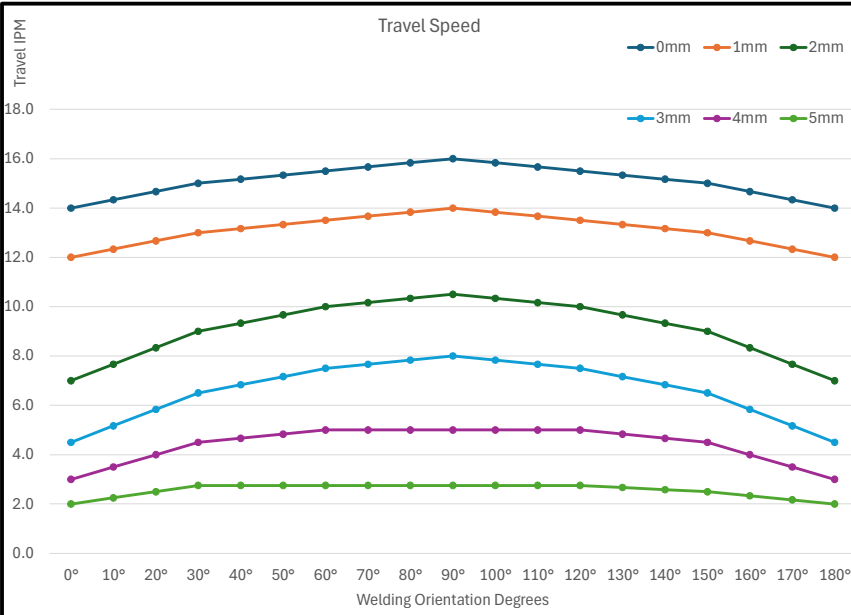
90°	Travel Speed	Torch Angle	Dynamic Correction
0mm	16	-5	-2
1mm	14	-5	-2
2mm	10.5	-10	-2
3mm	8	-15	-1
4mm	5	-15	0
5mm	2.75	-15	0

120°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15.5	0	-3
1mm	13.5	-5	-3
2mm	10	-10	-2
3mm	7.5	-15	-1
4mm	5	-15	0
5mm	2.75	-15	0

180°	Travel Speed	Torch Angle	Dynamic Correction
0mm	14	0	-5
1mm	12	0	-5
2mm	7	0	-4
3mm	4.5	0	-3
4mm	3	0	0
5mm	2	0	0

150°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15	0	-4
1mm	13	0	-4
2mm	9	-5	-3
3mm	6.5	-15	-2
4mm	4.5	-15	0
5mm	2.5	-15	0

Deposition Adjustment			CTWD	Arc Length	Weave Type	Weave Frequency	Weave Width	Weave Length	Weave Dwell
Gap	Whole Pipe	Lower Half							
0mm	0%	0%	-3	0	None	0	1.2	1.2	0
1mm	0%	0%	-3	0	None	0	1.2	1.2	0
2mm	0%	0%	-2	0	Circle	12	1.4	1.4	0
3mm	0%	0%	0	0	Circle	12	3.6	3.6	0
4mm	0%	0%	1	0	Circle	12	5.2	3.8	0
5mm	20%	10%	2	0	Circle	12	6.4	4.8	0



Cleaning / Grinding				
Root Gap	Grind Weld Pass	Travel Speed	Weave Amplitude	Frequency Hz
0mm	ON	100	3	2
1mm	ON	90	3	2
2mm	ON	90	3	2
3mm	ON	90	4	2
4mm	ON	90	4.5	2
5mm	ON	90	5	2

Notes	
Changes:	First attempt at open gap
Results:	

Weld Schedule Manager (Fill & Cap)



WPS No	J Prep TW1	Position	Fixed 5G - Down	Diameter (in)	48	Wall (mm)	15.8
Process	GTAW	Specification	API1104	Filler Wire	Lincoln Pipeliner ER70-S6 (1.2mm)	Shield Gas	Ar/C02 85/15
Bevel Design	J Prep	Material Grade	X70	Trade Name	Sure-arc	Completed by	Ben Hatch

0° Position	
Travel Speed	15
Torch Angle	0

30° Position	
Travel Speed	16
Torch Angle	-5

60° Position	
Travel Speed	18
Torch Angle	-10

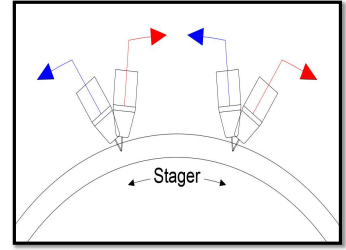
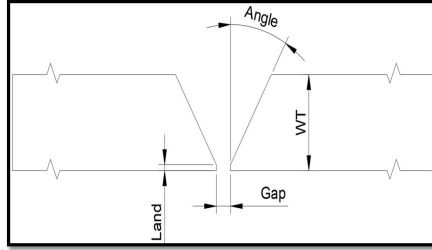
90° Position	
Travel Speed	20
Torch Angle	-10

120° Position	
Travel Speed	18
Torch Angle	-10

150° Position	
Travel Speed	14
Torch Angle	-10

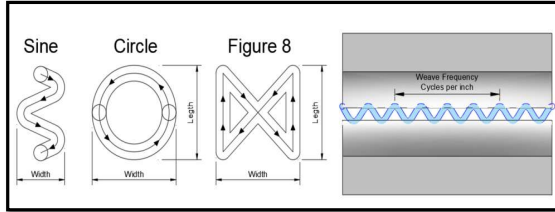
180° Position	
Travel Speed	10
Torch Angle	0

Wire Diameter	
1.2	
Direction of Travel	
Vertical Down	
Process	
Pulse	
Deposition Ratio	
12	
Shield Gas Flow (L/min)	
26	



Joint Geometry		Tolerance (+ / -)	
Wall Thickness (mm)	15.8	1	0.5
Land (mm)	1	0.5	0.5
Root Gap (mm)	3	1	1
Bevel Angle (°)	30	5	5
Hi / Low (mm)		1.5	1.5

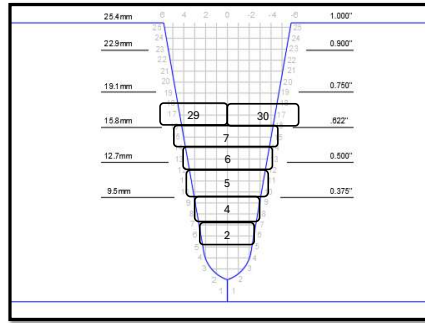
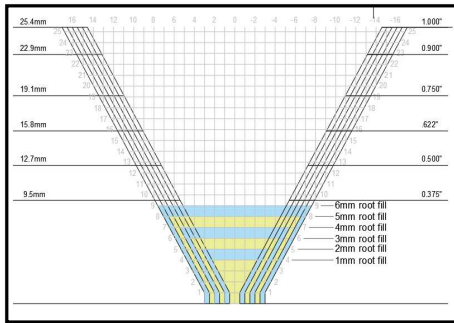
Stop / Start	
Stager the Stop-Start (mm)	30
Over run the next pass (mm)	12
Lead in at re-start (mm)	10



Weld Characteristics	
Deposition Efficacy	0.98
Pass Volume (mm2)	12.5
Min Cap Width (mm)	24
Root Depth (mm)	6
Total Passes (Inc root)	12

Total Arc Time (mins)	
28.8	
Wire Consumption (lb)	
9	
Wire Consumption (kg)	
4	

	Weave Adjustment				Weld Adjustment					Cleaning / Grinding					Comments	
	Weave Pattern	Width (mm)	Legth (mm)	Dwell (sec)	Frequency / inch	Deposition Adjustment	Travel Speed Adjustment	Pass Volume	Pass Time	Arc Legth	Grind Prior to Weld Pass	Grind After Weld Pass	Travel Speed (in/min)	Weave Amplitude		Frequency Hz
Hot Pass 0mm	Sine	1.5	2.0	0.0	12	-20%	-20%	10.6	2.9	2	OFF	ON	75	1	2	
	Sine	1.5	2.0	0.0	12	-10%	-10%	12.0	2.6	2	ON	OFF	75	1	2	
Hot Pass 1mm	Sine	1.5	2.0	0.0	12	-10%	-10%	12.0	2.6	2	ON	OFF	75	1	2	
	Sine	1.5	2.0	0.0	12	-10%	-10%	12.0	2.6	2	ON	OFF	75	1	2	
Hot Pass 2mm	Sine	1.5	2.0	0.0	12	-10%	-10%	12.0	2.6	2	ON	OFF	75	1	2	
	Sine	1.5	2.0	0.0	12	-10%	-10%	12.0	2.6	2	ON	OFF	75	1	2	
Hot Pass 3mm	Sine	1.5	2.0	0.0	12	0%	0%	13.3	2.3	2	ON	OFF	75	1	2	
	Sine	1.5	2.0	0.0	12	0%	0%	13.3	2.3	2	ON	OFF	75	1	2	
Hot Pass 4mm	Sine	1.5	2.0	0.0	12	0%	0%	13.3	2.3	2	ON	OFF	75	1	2	
	Sine	1.5	2.0	0.0	12	0%	0%	13.3	2.3	2	ON	OFF	75	1	2	
Hot Pass 5mm	Sine	1.5	2.0	0.0	12	0%	0%	13.3	2.3	3	ON	OFF	75	1	2	
	Sine	2.5	3.5	0.0	12	0%	0%	13.3	2.3	4	ON	ON	75	1	2	
Fill Passes	Sine	1.5	2.0	0.0	12	0%	0%	13.3	2.3	4	ON	ON	75	1	2	
Cap Passes	Sine	2.5	3.5	0.0	12	0%	0%	13.3	2.3	4	ON	ON	75	1	2	



Pass #	On/Off	Pass Type	0mm		1mm		2mm		3mm		4mm		5mm		Comments
			Vert offset	Horiz	Vert offset	Horiz Offset	Vert offset	Horiz Offset	Vert offset	Horiz Offset	Vert offset	Horiz Offset	Vert offset	Horiz Offset	
2	ON	Hot Pass	5.0	0.0	5.0	0.0	5.0	0.0	5.0	0.0	5.0	0.0	5.0	0.0	
3	OFF														
4	ON	Fill Pass	9.0	0.0	9.0	0.0	9.0	0.0	9.0	0.0	9.0	0.0	9.0	0.0	
5	ON	Fill Pass	12.0	0.0	12.0	0.0	12.0	0.0	12.0	0.0	12.0	0.0	12.0	0.0	
6	ON	Fill Pass	14.0	0.0	14.0	0.0	14.0	0.0	14.0	0.0	14.0	0.0	14.0	0.0	
7	ON	Fill Pass	16.0	0.0	16.0	0.0	16.0	0.0	16.0	0.0	16.0	0.0	16.0	0.0	
8	OFF														
9	OFF														
10	OFF														
11	OFF														
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25	OFF														
26	OFF														
27.0	OFF														
28.0	OFF														
29.0	ON	Cap Pass	13.0	3.0	13.0	3.0	18.0	3.0	13.0	3.0	13.0	3.0	13.0	3.0	
30.0	ON	Cap Pass	13.0	-3.0	13.0	-3.0	18.0	-3.0	13.0	-3.0	13.0	-3.0	13.0	-3.0	

Notes

Changes: We made the bottom weld with no torch angle.

Results: New Baseline. It was very slightly flatter across the passes from not dragging.

Suggestions: